Advanced Mechanics of Materials
Fall 2026

Instructor Information

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Course Information

Section Information: AE 6304-001/002; ME 6304-001/002
Course Description:
Course Title: Advanced Mechanics of Materials
This graduate-level course will cover the calculation of stresses and strains in a body that experiences hyperelastic, viscoelastic, and plastic deformation. Offered as AE 6304 and ME 6304. Credit will be granted only once.
Prerequisite: AE 5339, ME 5339, or instructor consent.

Time and Place of Class Meetings

Tuesdays and Thursdays – 3.30 pm to 4.50 pm.
This course operates on Central Time. All times listed for class meeting times, exams, and assignment deadlines are in Central Time (CT).

Classroom/Lecture Recording Policy

Faculty maintain the academic right to determine whether students are permitted to record classroom and online lectures. Recordings of classroom lectures, if permitted by the instructor or pursuant to an ADA accommodation, may only be used for academic purposes related to the specific course. They may not be used for commercial purposes or shared with non-course participants except in connection with a legal proceeding.

Recording of classroom and online lectures in this course is not allowed.

Student Learning Outcomes

After the completion of this course, students should be able to:

- Describe the elastic, hyperelastic, plastic and viscoelastic behavior of materials from their stress-strain relations.
- Solve simple boundary value/initial value problems using elastic, elasto-plastic, viscoelastic, and hyperelastic properties.

- Derive differential constitutive relations for viscoelastic materials using phenomenological models.
- Explain the theoretical foundations of hyperelasticity and distinguish hyperelastic behavior from linear elasticity, plasticity, and viscoelasticity under finite deformations.
- Analyze and compare classical hyperelastic constitutive models (e.g., Neo-Hookean, Mooney–Rivlin, and Arruda–Boyce) and assess their suitability for different classes of engineering materials.
- Determine the constitutive relation between a mechanical stress and an electrical voltage in piezoelectric solids.

Course Materials & Technology

Textbook Information

None. All necessary materials will be supplied by the instructor.

Suggested Reading:

1) Theory of Elasticity (3rd Edition) by S.P. Timoshenko and J. N Goodier
2) Theory of Viscoelasticity (2nd Edition) by Richard M. Christenson
3) Theory of Plasticity (3rd Edition) by J. Chakrabarty
4) Hyperelasticity Primer (2nd Edition) by Robert Hackett
5) Mechanics of Soft Materials (2nd Edition) by Konostantin Volokh
6) Special Topics in the Theory of Piezoelectricity by Jiashi Yang (Editor)

Technology & Equipment Requirements

Visit the OIT Services page for a list of Applications and Software available through UTA.

Visit the UTA Libraries Technology page for a list of items that can be checked out or used at the library.

Assignments & Exams

Descriptions of major assignments and examinations:

- There will be homework assignments, a project, one midterm exam, and one final exam.
- Details about the project will be discussed in class.

- **Homework:** Assigned during the Tuesday classes and due dates are defined in the assignment (tentatively due in 2 weeks).
- **Midterm:** Tentative schedule - October 6, 2026 (combination of in-class and take-home)
- **Final Project Due and Presentation:** December 1, 2026

- **Final Exam:** December 8, 2026, 2.00 pm to 4.30 pm. (combination of in-class and take-home)

Grading Information

- **Grading Policy**: Final numerical grade for each student will be determined based on his/her total earned points on a scale from 0 to 100. The total earned points for each student will be computed by adding points obtained by the student from the midterm exam, homework assignments, and the final exam.

- **Point Distributions:**

Homework + Project	30
Midterm	35
Final Exam	35
Total points	100

Final Grade Calculations

The final letter grade for each student will be determined from his/her final numerical grade according to the following ranges:

Tentative Grading Scale	Letter Grade
90 – 100	A
80 – 89	B
70 – 79	C
60 – 69	D
00 – 59	F

Late Work Policy

- Homework turned in late (after 5.00 pm on the due date) will receive a 20% penalty per class until the solution is posted. No late homework will be accepted after the solution is posted.

Make-Up Exams Policy

- UTA regulations permitting, missed midterms and/or final exams can only be rescheduled when missed due to major health problems or circumstances beyond the student's control. With instructor discretion, students will be required to reschedule the missed exams at the earliest time possible.

Extra Credit Policy

- None

Grade Grievance Policy

- For the class, any questions or review requests of graded papers must be made within 5 business days. Any appeal of a grade in this course must follow the procedures and deadlines for grade-related grievances as published in the current University Catalog: Grades and Grading Policies.

Course & University Policies

Students should review the University Class Attendance Policies on the Class Attendance Policies page. For this class, students are highly encouraged to attend classes and engage in discussions. Attendance will be taken daily but is not mandatory.

Institutional Policies

UTA students should review the University Catalog and the Syllabus Institutional Policies page for institutional policies and contact the specific office with any questions. The institutional information includes the following policies, among others:

- Drop Policy
- Disability Accommodations
- Academic Integrity
- Electronic Communication

UTA Honor Code

UTA students are expected to adhere to and observe standards of conduct compatible with the University's functions as an educational institution and live by the University of Texas at Arlington's Honor Code. It is the policy of The University of Texas at Arlington to uphold and support standards of personal honesty and integrity for all students consistent with the goals of a community of scholars and students seeking knowledge and responsibility.

Student Support Services

Student Services Page

The Student Services page provides links to many resources available to UTA students, including:

- Academic Success
- Counseling and Psychological Services (CAPS)
- Health Services
- Students with Disabilities
- Veteran Services

Students are also encouraged to check out Career Center resources to enhance their career-readiness, find student employment, search for internships, and more. We encourage Major

Exploration and the use of Experiential Major Maps to keep students on track for graduation. Refer to the Graduation Help Desk for more details.

Accessibility of Course Materials

Some course materials, such as PDFs of musical scores, technical drawings, graphs, blueprints, design plans, or artworks (common in fields like drawing, painting, or construction drafting), may not fully comply with all Web Content Accessibility Guidelines (WCAG) requirements.

The University of Texas at Arlington is dedicated to ensuring all students have equal access to information. If you experience any accessibility barriers with course materials, please know that accommodations are available. You can get assistance through the Student Access and Resource (SAR) Center or by contacting your instructor directly. Please don't hesitate to reach out if you need help.

Online Academic Success Guide

Visit the Online Academic Success Guide to explore a list of helpful tips and resources to help you succeed in your online journey.

Course Schedule

Lecture	Mechanics Topic	Topic
1	Statics and Solid Mechanics	Review of Mechanics of Material - I
2		Review of Mechanics of Material - II
3	Linear Elasticity	Review of Linear Elasticity
4		Strain, Stress Measures
5		Consitutive Relations
6		Review of 2D Boundary Value problems
7		Review of 2D Boundary Value problems
8	Reserve	
9	Viscoelasticity	Foundation of Viscoelasticity
10		Differential Constitutive Equations
11		Hereditary Integral Representations of Stress and Strain
12		Elementary Viscoelastic Stress Analysis for Bars and Beams
13		Viscoelastic Stress Analysis in Two and Three Dimensions - I
14		Viscoelastic Stress Analysis in Two and Three Dimensions - II
15	Midterm Exam (tentative)	
16	Plasticity	Foundation of Plasticity
17		Elastoplastic Bending/Torsion
18		Elastoplastic Boundary Value Problems - I
19		Elastoplastic Boundary Value Problems - II
20		Plasane Problems in Plastic Flow and Collapse -I
21		Plane Problems in Plastic Flow and Collapse -II
22	Hyperelasticity	Revisit Finite Elasticiy and Strain Measures
23		Strain Energy Functions and Polar Decompositions
24		Hyperelastic Constitutive Theory
25		Hyperelastic Constitutive Theory
26	Reserve	
27	Piezoelasticity	Theory and Mathematical Foundation
28		2D Piezoelectric Boundary Value Problem - I
29		2D Piezoelectric Boundary Value Problem -II
30	Project Presentation	

As the instructor for this course, I reserve the right to adjust this schedule in any way that serves the educational needs of the students enrolled in this course. –Ashfaq Adnan